

Characteristics of historical Compound Flash Droughts and

 neuromatch ISP

 Search CTRL K



AUTHORS

Vega-Jacome, F. 

Wu, J. 

Singh, M. 

AFFILIATIONS

Institute of Environmental Science and Geography,
University of Potsdam, Potsdam, Germany

Institute for Biology, Humboldt University of Berlin,
Germany

Alliance Bioversity-CIAT

Abstract

Extreme climate events such as heatwaves, flash droughts, and compound events pose increasing risks to society under a changing climate. In this study, the spatiotemporal characteristics of Compound Flash Drought Heatwave (CFDHW) events were studied using high-resolution gridded datasets of precipitation, temperature, and potential evapotranspiration across Peru from 1981 to 2015.

Compound events were defined as the co-occurrence of heatwaves and flash droughts in the same spatial location. Our results reveal strong geographic contrasts in the distribution of the extreme events. Heatwaves were frequent and persistent along the coast and in the northern Amazon basin. In contrast, flash droughts were more widespread and lasted longer in the central and southern parts of the Pacific basin. CFDHW events were less frequent by nature of detection, but showed clear clustering in coastal and northern Amazon regions, in accordance with the heatwaves. However, CFDHW events lasted longer at the coast and in the central Amazon basin. Interannual variability indicates that the severity of heatwaves and compound events peaked mainly during major El Niño episodes (e.g., 1982–83 and 1997–98), whereas flash drought severity showed several peaks, especially after 2000.

Keywords: Compound flash drought and heatwave events, climate variability, Extreme events, Peru

Introduction

Compound climate events are defined as combinations of multiple drivers and/or hazards that jointly contribute to societal or environmental risk ([Zscheischler *et al.* \(2018\)](#)). The evolution of droughts and heatwaves is significantly influenced by positive land–atmosphere feedbacks between these extreme events ([Marengo *et al.* \(2025\)](#); [Miralles *et al.* \(2019\)](#); [Rouges *et al.* \(2023\)](#)), and should be analyzed as compound events rather than individual events ([Zscheischler & Seneviratne \(2017\)](#)). A core challenge is to analyze both hazards on comparable timescales, since heatwaves typically last for several days to a few weeks ([Miralles *et al.* \(2019\)](#)), whereas classical droughts develop at the scale of months ([Stagge *et al.* \(2015\)](#)), making it hard to understand the combined effects of co-occurring hazards. Here, we consider a special form of drought characterized by rapid onset and short duration referred to as flash drought ([Oikonomou *et al.* \(2020\)](#); [Lisonbee *et al.* \(2021\)](#)). In Peru, only a handful of reports on either heatwaves or droughts are available ([Gálvez \(2018\)](#); [Quispe *et al.* \(2015\)](#), [Vega-Jacome & Lavado-Casimiro \(2015\)](#); [Zubieta *et al.* \(2021\)](#); [Ruedas *et al.* \(2023\)](#)), and none of them has analyzed droughts at a comparable timescale as heatwaves. Regarding compound drought and heatwave events, only particular past events were analyzed. Here, we aim to characterize the spatiotemporal pattern of the compound flash drought and heatwave (CFDHW) events during the climatological period from 1981 to 2015 in Peru.

Material and methodology

Study area

Peru, located in the central and western region of South America, extends from 0°02' N to 18°21' S in latitude and 81°19' W to 68°39' W in longitude. From east to west, the Andes divide the territory into a desert coastal region to the west and the Amazon basin to the east, with elevations ranging from sea level to 6,768 m at Huascarán glacier. The complex topography of Andes induces strong hydroclimatic variability across multiple spatial and temporal scales ([Arias *et al.* \(2021\)](#); [Espinoza *et al.* \(2015\)](#); [Casimiro *et al.* \(2012\)](#)). Due to its complex hydroclimatology, precipitation distribution varies significantly both latitudinally (north-south) and longitudinally (east-west), shaping diverse climatic conditions. These range from the arid and warm coastal region to the temperate, frigid, and polar inter-Andean valleys, and the warm and humid Amazon rainforest.

Data

To identify heatwaves and flash droughts, we used high-resolution gridded long-term historical daily climate data from 1981 to 2015, including precipitation, maximum and minimum temperature, and potential evapotranspiration data ([Table 1](#)). Since the variables were obtained at different spatial resolutions, regridding and clipping were performed to standardize the grid resolution and the spatial extent. Additionally, a shape file of the country was applied to exclude data outside Peruvian territory.

Methodology and Tools

First, heatwaves and flash droughts were identified separately at a comparable timescale. For heatwave detection, both daily maximum and minimum temperatures were smoothed with a kernel of the size of five days. Heatwaves were then detected for three or more consecutive days with the smoothed maximum and minimum temperatures above the corresponding 90th percentile of the reference period (1981–2010, [Fu & Wang \(2022\)](#)).

Table 1: Description of climate products used in the analysis

Climate variable	Name of product	Time period	Spatial resolution	Temporal resolution	Reference
Precipitation	RAIN4PE - Rain for Peru and Ecuador	1981–2015	0.1°	daily	Fernandez-Palomino <i>et al.</i> (2022)

Temperature (Tmax, Tmin)	SENAMHI - PISCOT v1.2	1981– 2020	0.01°	daily	Huerta <i>et al.</i> (2023)
Potential Evapo- transpiration (PET)	Penman-Monteith ETo dataset (PISCOeo_pm)	1981– 2016	~ 1 km	daily	Huerta <i>et al.</i> (2022)

To identify flash droughts, the standardized precipitation evapotranspiration index was estimated with a window size of five days (pentad-SPEI) using the daily precipitation data from the RAIN4PE product and daily potential evapotranspiration (PET) data. A large negative SPEI value indicates dry conditions, while a large positive SPEI indicates wet conditions. Here, we used a daily pentad-SPEI, which is more comparable to the smoothed daily temperature data. Adapted from [Fu & Wang \(2022\)](#) to our daily pentad-SPEI, a flash drought is detected when:

- a daily pentad-SPEI drops below the 40th percentile (onset threshold) and continues to drop below the 20th percentile (detection threshold),
- from the onset to the minimum, no more than one pentad has a decline rate < 5 percentiles,
- a minimum duration of 3 pentads, i.e. 15 days, is observed before the daily pentad-SPEI recovers to >=20th percentile (offset threshold),
- a minimum duration of 1 pentad, i.e. 5 days, is observed with the daily pentad-SPEI <= 20th percentile, and
- the mean daily pentad-SPEI from the onset to the offset is <= 30th percentile.

The CFDHW events were defined as the co-occurrence of heatwaves and flash droughts ([Figure 1](#) a). Heatwaves, flash droughts, and compound events were all characterized in terms of the number of events, annual frequency, and mean duration. Additionally, we quantify the severity of each extreme event. Heatwave severity is the cumulative exceedance of smoothed maximum temperature above its 90th percentile threshold ([Tripathy & Mishra \(2023\)](#)); flash drought severity is the cumulative deficit of pentad-SPEI below the 40th percentile; compound severity follows [Alizadeh *et al.* \(2020\)](#), computed as the sum of –pentad-SPEI × normalized Tmax over the overlap window.

Result

The spatial patterns of extreme climate events revealed clear geographic contrasts across Peru ([Figure 1](#) c). Heatwaves were more frequent and had longer durations along the coast.

In the northern Amazon basin, although the heatwaves were frequent, their durations were much shorter than in the coastal regions. This suggests persistent and recurrent thermal stress in these areas, likely influenced by arid coastal conditions and ocean–atmosphere interactions. In contrast, the Andes region (highlands) exhibited comparatively fewer and shorter heatwaves, which could be explained by the regulating effect of the high altitudes and the glaciers on the minimum temperatures. Flash droughts displayed a broader and more spatially uniform pattern than heatwaves, with higher frequency along the Pacific and Titicaca basin, and higher durations mainly in the center and the south of the Pacific basin. Flash droughts lasted substantially longer (16–26 days) than heatwaves (3–17 days), indicating sustained moisture deficits once drought conditions are triggered. The southern regions, already prone to water scarcity, were especially vulnerable to repeated drought stress. Compound FDHW events, defined by the co-occurrence of heatwaves and flash droughts, were less frequent overall, but displayed clustering in coastal and parts of the north Amazon basin of Peru. In central Peru, compound events lasted longer but occurred less frequently compared to other parts of Peru.

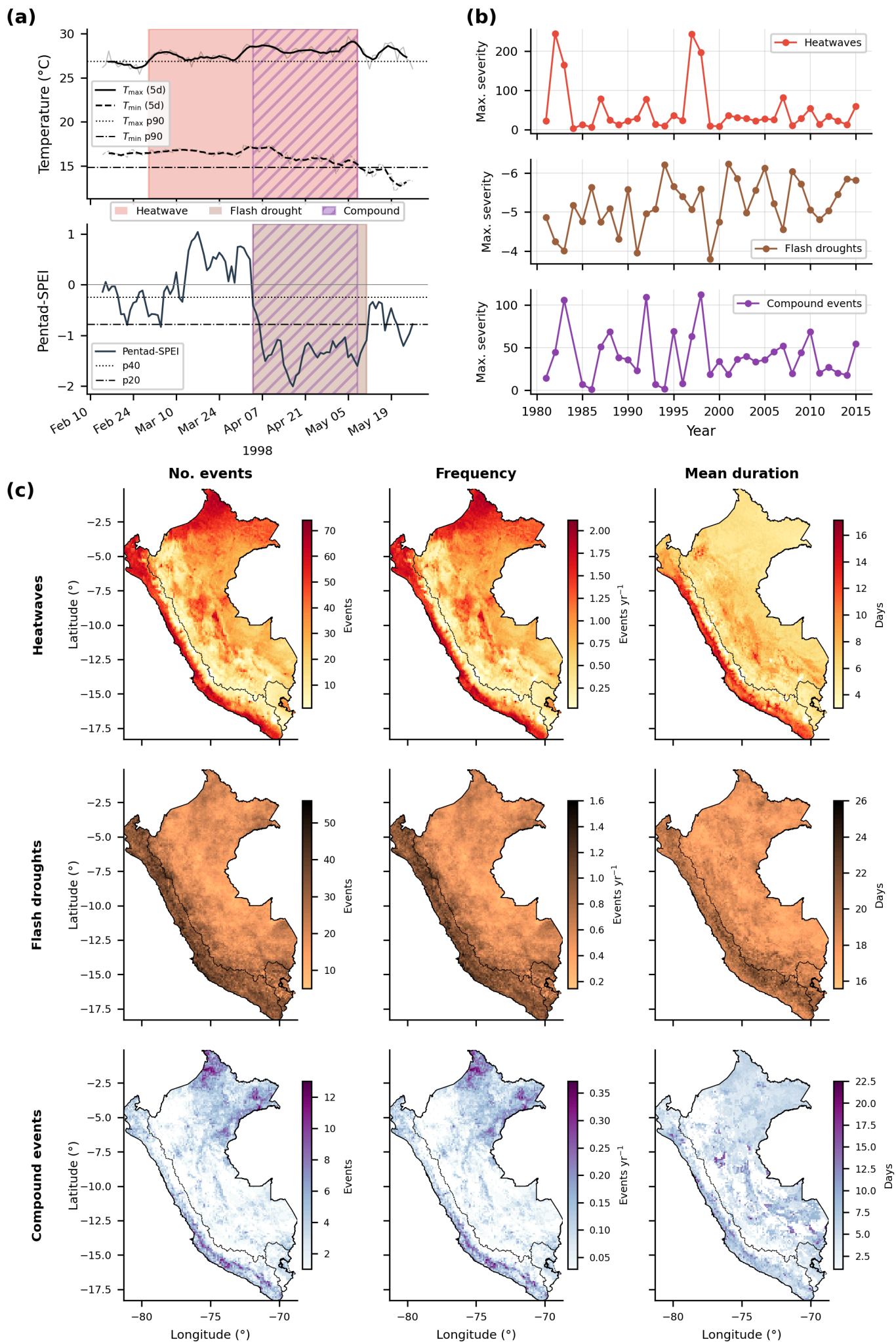


Figure 1: Compound flash droughts and heatwaves across major hydrological basins in Peru (1981–2015).

a. Example detection of a compound event at a single pixel during 1998, showing daily maximum and minimum temperatures smoothed using a moving window of 5 days (top) and daily pentad-SPEI (bottom). Heatwaves (red shading) and flash droughts (brown shading) are detected using the respective thresholds expressed as the N-th percentiles (horizontal lines) in accordance with Fu & Wang (2022); compound events (purple hatched shading) are periods of temporal overlap between concurrent heatwave and flash drought events. .

b. Annual maximum severity of heatwaves (top, red), flash droughts (middle, brown; y-axis inverted), and compound events (bottom, purple) across Peru over the study period.

c. Spatial distribution of event characteristics across Peru: total number of events (left), mean annual frequency (center), and mean duration in days (right) for heatwaves (top row), flash droughts (middle row), and compound events (bottom row). Black solid lines show the national boundary; dashed lines delineate the three major hydrographic watersheds (Pacific, Atlantic, and Titicaca).

Next, we examined the temporal variability of extreme climate events across the reporting period (1981–2015). The highest severity of extreme events showed pronounced interannual variability across all event types ([Figure.1 b](#)). Flash drought severity remained relatively stable within a narrow range of negative values, exhibiting frequent year-to-year fluctuations with a visible positive long-term trend suggesting increasingly drier conditions overall. Notably, no consistent signal of enhanced severity was observed even during major El-Niño years (1982–83, and 1997–98). El-Niño events assert contrasting hydroclimatic effects along the longitudinal axis of Peru, i.e., while northern regions typically experience increased precipitation, southern regions often undergo drying. The maximal severity alone was likely insufficient to reflect the effect of El-Niño on flash droughts. In contrast, heatwave severity shows stronger interannual variability, with the highest peaks occurring during major El-Niño episodes (particularly in 1982–83, and 1997–98). This pattern suggests prolonged and intensified warm conditions across large parts of the country during these periods, reflected in both increased amplitude and duration of heatwaves. Compound events also show episodic behaviour, with several pronounced peaks that generally coincide with moderate to high severity heatwave years. This suggests that compound FDHW extremes are primarily driven by periods of enhanced thermal stress, with concurrent land–atmosphere feedbacks likely amplifying drought conditions during warm extremes. Like in the case of the compound FDHW event of 1998 over the south of the Pacific basin ([Figure.1 a](#)), where the heatwave event started at the beginning of March, followed by a flash drought starting in April, lasting until May.

Conclusion

This study characterises the spatio-temporal dynamics of compound flash drought–heatwave (CFDHW) events across Peru, revealing strong regional contrasts and distinct temporal variability. Heatwaves were more frequent along the Pacific coast and in the

northern Amazon basin, whereas flash droughts occurred more frequently and lasted longer in the Pacific and Titicaca basins. Consequently, compound events were most prevalent along the southern Pacific coast and in the northern Amazon basin. The temporal variability revealed that the most severe heatwaves occurred during El Niño years, while flash drought severity exhibited high inter-annual variability with an overall increasing trend. CFDHW severity exhibited greater variability before 2000, with pronounced peaks linked to major El Niño events. These CFDHW events typically evolved through an initial phase of intensified heatwaves, followed by the onset of flash drought conditions, and are modulated by land-atmosphere feedback processes.

Methodologically, the use of a pentad-scale framework enabled a more precise identification of the timing and co-occurrence of heatwaves and flash droughts, addressing limitations in previous approaches (add ref.). To the best of our knowledge, this is the first study in Peru to explicitly account for the temporal alignment of these extreme climate events. Focusing on co-occurring extremes, i.e., by examining periods experiencing both extremes - called CFDHW events, provides a clearer understanding of their interaction and impacts across key sectors (agriculture, industry, and urban area), where combined heat and dryness can strongly affect crops, water availability, infrastructure, and living conditions. Risks are greater along the already arid coastal region, where prolonged heatwaves and persistent moisture deficits could place sustained pressure on coastal agriculture, urban water supply, and energy systems. In southern regions, longer flash droughts intensify existing water stress, while in the northern Amazon, shorter but recurring heat extremes may increase the risk of forest fires in this region. The Andes, although relatively less exposed to prolonged heatwaves, remain sensitive due to their dependence on cryospheric and high-altitude climate regulation. Together, these spatial contrasts highlight that compound extremes do not impact Peru uniformly, but instead pose region-specific risks that require targeted adaptation and management strategies.

Acknowledgments

This work was supported by the Impact Scholars Program. We acknowledge the contributions of viviana, Ayuk[former team members, teaching assistants, or mentors whose involvement does not meet the criteria of any authorship role].

References

[Show All](#)

1. Zscheischler, J., Westra, S., Hurk, B. J. J. M. V. D., Seneviratne, S. I., Ward, P. J., Pitman, A., Aghakouchak, A., Bresch, D. N., Leonard, M., Wahl, T., & Zhang, X. (2018). Future climate risk from compound events. *Nature Climate Change*, 8(6), 469–477. [10.1038/S41558-018-0156-3](https://doi.org/10.1038/S41558-018-0156-3)

2. Marengo, J. A., Costa, M. C., Cunha, A. P., Espinoza, J. C., Jimenez, J. C., Libonati, R., Miranda, V., Trigo, I. F., Sierra, J. P., Geirinhas, J. L., Ramos, A. M., Skansi, M., Molina-Carpio, J., & Salinas, R. (2025). Climatological patterns of heatwaves during winter and spring 2023 and trends for the period 1979–2023 in central South America. *Frontiers in Climate*, 7. [10.3389/fclim.2025.1529082](#)
3. Miralles, D. G., Gentile, P., Seneviratne, S. I., & Teuling, A. J. (2019). Land–atmospheric feedbacks during droughts and heatwaves: state of the science and current challenges. *Annals of the New York Academy of Sciences*, 1436(1), 19–35. [10.1111/nyas.13912](#)
4. Rouges, E., Ferranti, L., Kantz, H., & Pappenberger, F. (2023). European heatwaves: Link to large-scale circulation patterns and intraseasonal drivers. *International Journal of Climatology*, 43(7), 3189–3209. [10.1002/joc.8024](#)
5. Zscheischler, J., & Seneviratne, S. I. (2017). *Dependence of drivers affects risks associated with compound events* [Techreport]. <https://www.science.org>
6. Stagge, J. H., Tallaksen, L. M., Gudmundsson, L., Loon, A. F. V., & Stahl, K. (2015). Candidate Distributions for Climatological Drought Indices (SPI and SPEI). *International Journal of Climatology*, 35(13), 4027–4040. [10.1002/joc.4267](#)
7. Oikonomou, P. D., Karavitis, C. A., Tsesmelis, D. E., Kolokytha, E., & Maia, R. (2020). Drought Characteristics Assessment in Europe over the Past 50 Years. *Water Resources Management*, 34(15), 4757–4772. [10.1007/s11269-020-02688-0](#)
8. Lisonbee, J., Woloszyn, M., & Skumanich, M. (2021). Making sense of flash drought: definitions, indicators, and where we go from here. *Journal of Applied and Service Climatology*, 2021(1), 1–19. [10.46275/JOASC.2021.02.001](#)
9. Gálvez, F. C. C. (2018). *Olas de Calor en la Amazonia Peruana* (pp. 1–149) [Phdthesis]. UNALM.
10. Quispe, K., Luis, V., Lozano, A., Gutiérrez, N. Q., Galvez, F. C., & Técnica, N. (2015). *IDENTIFICACIÓN DE EVENTOS DE “OLAS DE CALOR” EN LA AMAZONÍA PERUANA* (pp. 1–19) [Techreport]. SENAMHI-PERU. www.senamhi.gob.pe
11. Vega-Jacome, F., & Lavado-Casimiro, W. (2015). *Regionalización y caracterización de sequías en el Peru* (pp. 1–33) [Techreport]. Peruvian Meteorological Service.
12. Zubieta, R., Molina-Carpio, J., Laqui, W., Sulca, J., & Ilbay, M. (2021). Comparative analysis of climate change impacts on meteorological, hydrological, and agricultural droughts in the lake titicaca basin. *Water (Switzerland)*, 13(2). [10.3390/w13020175](#)
13. Ruedas, D. P. R. A., Guerra, L. S., Popli, K., & Madaki, S. G. (2023). Spatio-Temporal Drought Assessment Using Standardized Precipitation Evapotranspiration Index (SPEI) over Mantaro Valley, Peru. *Revista de Investigaciones Altoandinas - Journal of High Andean Research*, 25(3), 159–170. [10.18271/ria.2023.525](#)
14. Arias, P. A., Garreaud, R., Poveda, G., Espinoza, J. C., Molina-Carpio, J., Masiokas, M., Viale, M., Scaff, L., & van Oevelen, P. J. (2021). Hydroclimate of the Andes Part II: Hydroclimate Variability and Sub-Continental Patterns. In *Frontiers in Earth Science* (Vol. 8). Frontiers Media S.A. [10.3389/feart.2020.505467](#)
15. Espinoza, J. C., Chavez, S., Ronchail, J., Junquas, C., Takahashi, K., & Lavado, W. (2015). Rainfall hotspots over the southern tropical Andes: Spatial distribution, rainfall intensity, and relations with large-scale atmospheric circulation. *Water Resources Research*, 51(5), 3459–3475. [10.1002/2014WR016273](#)

Show all 22 references